Git & GitHub Assignment – Screenshots

Repository: github.com/yogesh0405/DevopsCourseTask

Task 1: Repository Setup & First Branch

1. Create a new GitHub repository.

2. Clone the repository using SSH (generate an SSH key if needed, and add the public key to GitHub).

```
git clone git@github.com:yogesh0405/DevopsCourseTask.git
```

```
yogesh@Yogeshs-MacBook-Air CI_Cd % git remote add origin https://github.com/yogesh0405/DevopsCourseTask.git
```

Adding the remote origin (git remote add origin)

3. Create a new branch named after your username.

```
yogesh@Yogeshs-MacBook-Air CI_Cd % git switch -c yogesh
Switched to a new branch 'yogesh'
```

Creating the branch: git switch -c yogesh

4. Add your Flask project files to this branch.

```
yogesh@Yogeshs-MacBook-Air CI_Cd % git status
On branch yogesh

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .dockerignore
        LinuxFundamentals/
        README.md
        backend/
        compose.yaml
        frontend/
```

git status showing the project files ready to be added

5. Commit the changes and merge the branch into main.

```
yogesh@Yogeshs-MacBook-Air CI_Cd % git commit -m "initial commit"
[main (root-commit) 2fe9f9b] initial commit
 Committer: Yogesh <yogesh@Yogeshs-MacBook-Air.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 599 files changed, 65502 insertions(+)
 create mode 100644 .dockerignore
 create mode 100644 LinuxFundamentals/Linux_Commands_Report.pdf
 create mode 100644 README.md
 create mode 100644 backend/Dockerfile
 create mode 100644 backend/app.py
 create mode 100644 backend/requirements.txt
 create mode 100644 compose.yaml
 create mode 100644 frontend/Dockerfile
 create mode 100644 frontend/node_modules/.package-lock.json
 create mode 100644 frontend/node_modules/accepts/HISTORY.md
```

Initial commit of the project files

```
yogesh@Yogeshs-MacBook-Air CI_Cd % git switch -c yogesh
Switched to a new branch 'yogesh'
yogesh@Yogeshs-MacBook-Air CI_Cd % git branch
  main
* yogesh
yogesh@Yogeshs-MacBook-Air CI_Cd % git add .
yogesh@Yogeshs-MacBook-Air CI_Cd % git commit -m "backgroud clr change"
[yogesh 7ccd794] backgroud clr change
 Committer: Yogesh <yogesh@Yogeshs-MacBook-Air.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

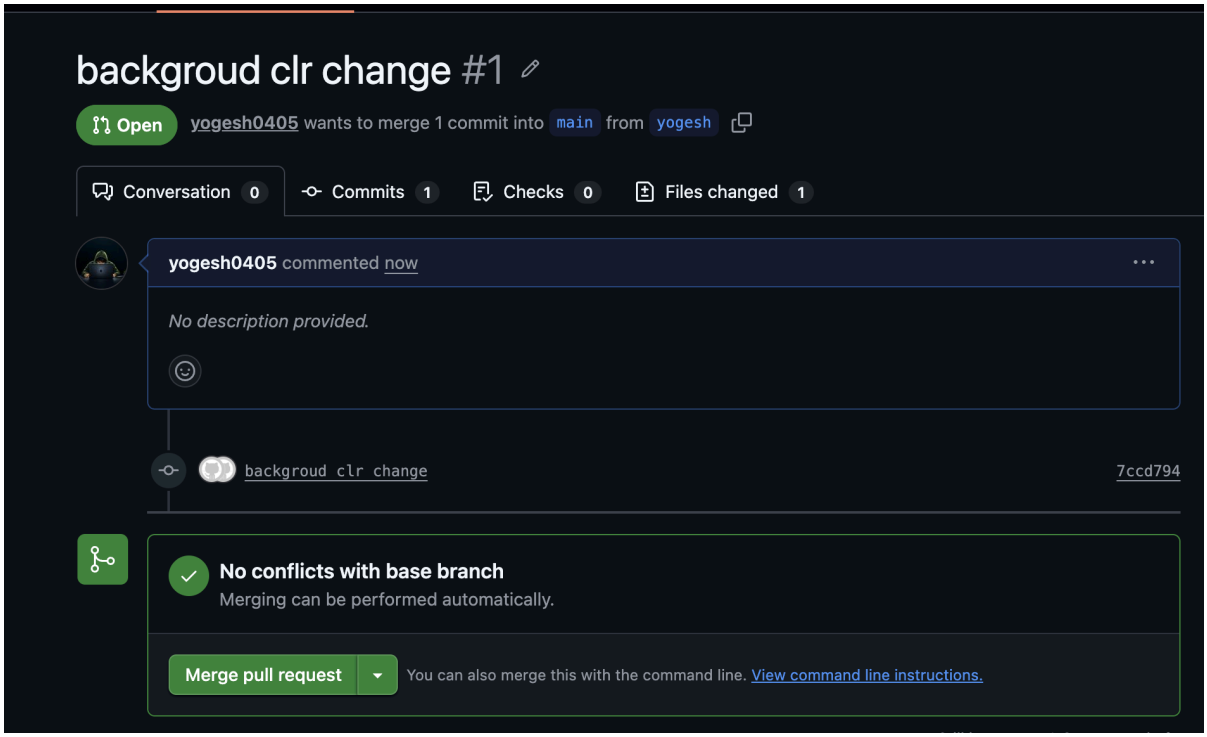
    git commit --amend --reset-author

 1 file changed, 1 insertion(+), 1 deletion(-)
yogesh@Yogeshs-MacBook-Air CI_Cd % git push -u origin yogesh
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
```

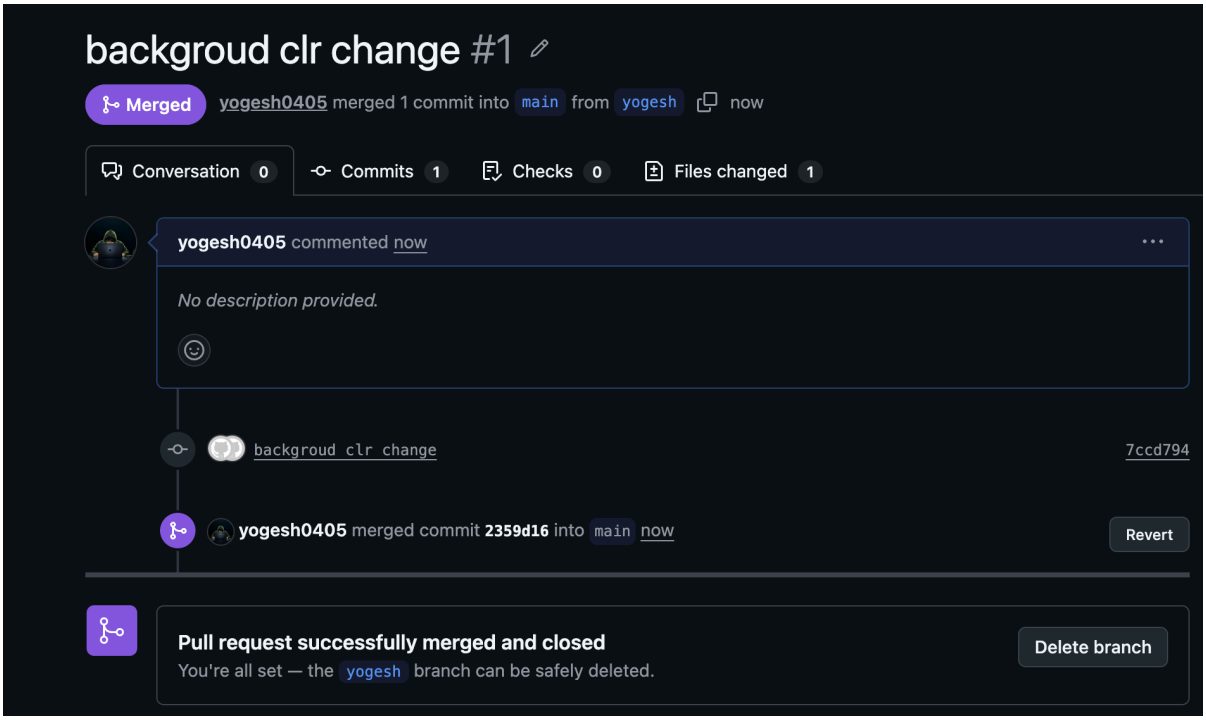
Branch yogesh: add, commit "backgroud clr change" and push

```
yogesh@Yogeshs-MacBook-Air CI_Cd % git push -u origin yogesh
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 423 bytes | 423.00 KiB/s, done.
Total 5 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 4 local objects.
remote:
remote: Create a pull request for 'yogesh' on GitHub by visiting:
remote:      https://github.com/yogesh0405/DevopsCourseTask/pull/new/yogesh
remote:
To https://github.com/yogesh0405/DevopsCourseTask.git
 * [new branch]      yogesh -> yogesh
branch 'yogesh' set up to track 'origin/yogesh'.
yogesh@Yogeshs-MacBook-Air CI_Cd %
```

Pushing the branch: git push -u origin yogesh



Pull request #1 open, no conflicts, ready to merge



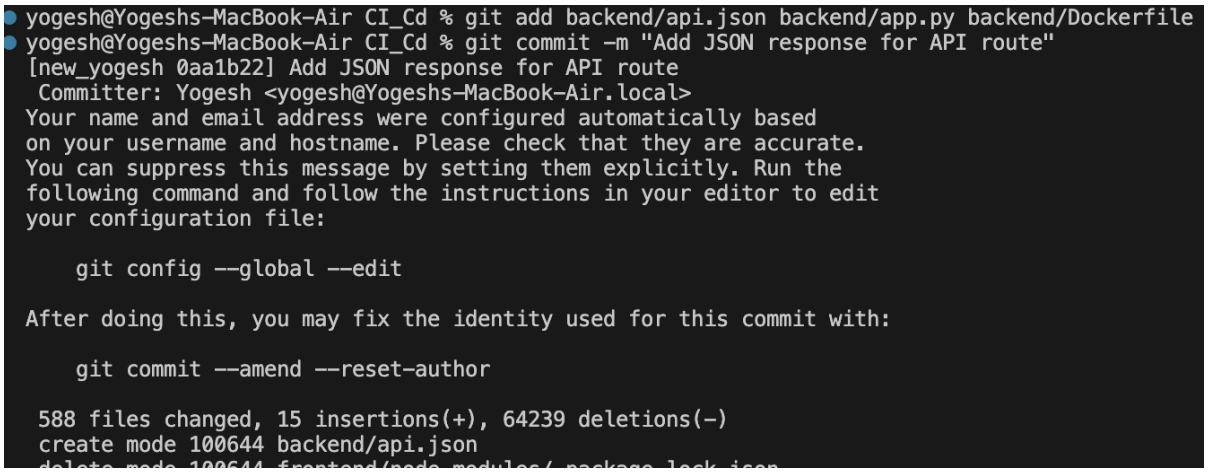
Pull request #1 merged into main

Task 2: Update JSON & Resolve Conflicts

6. Create a new branch named _new (e.g., yogesh_new).

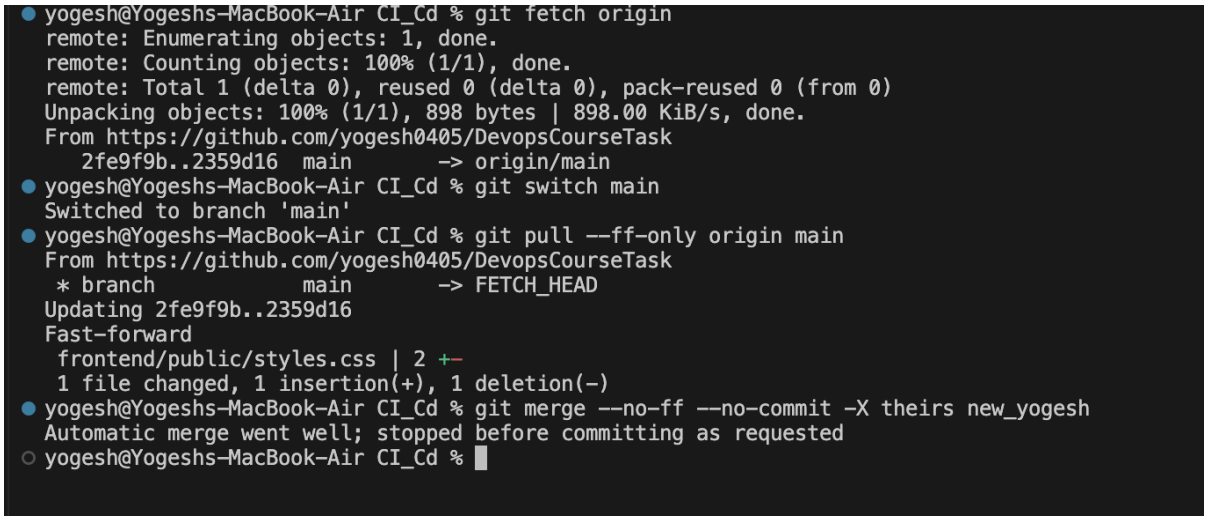
```
git switch main
git switch -c new_yogesh
```

7. Update the content of the JSON file used for the /api route in this branch.



Committing the JSON change on the new_yogesh branch

8. Merge the _new branch into the main branch.



Fetching and updating main, then merging new_yogesh with -X theirs

9. If there are conflicts, resolve them by accepting the changes from the _new branch.

```
git checkout --theirs backend/api.json
```

10. Add the resolved changes to staging, commit them, and push to the remote repository.

```
git add .
git commit -m "Merge _new into main"
git push origin main
```

Task 3: Parallel Feature Development

11. Branch Creation: Create two branches, master_1 and master_2, from main.

```
yogesh@Yogeshs-MacBook-Air CI_Cd % git switch main
Already on 'main'
yogesh@Yogeshs-MacBook-Air CI_Cd % git switch -c master_1
Switched to a new branch 'master_1'
yogesh@Yogeshs-MacBook-Air CI_Cd % git switch main
Switched to branch 'main'
yogesh@Yogeshs-MacBook-Air CI_Cd % git switch -c master_2
Switched to a new branch 'master_2'
yogesh@Yogeshs-MacBook-Air CI_Cd %
```

Creating master_1 and master_2 from main

12. Feature Development in master_1: Create a To-Do Page in the frontend with a form containing Item Name and Item Description.

```
yogesh@Yogeshs-MacBook-Air CI_Cd % git switch main
yogesh@Yogeshs-MacBook-Air CI_Cd % git switch -c master_2
Switched to a new branch 'master_2'
yogesh@Yogeshs-MacBook-Air CI_Cd % git switch master_1
Switched to branch 'master_1'
yogesh@Yogeshs-MacBook-Air CI_Cd % git branch
  main
* master_1
  master_2
  new_yogesh
  yogesh
yogesh@Yogeshs-MacBook-Air CI_Cd % gut status
zsh: command not found: gut
yogesh@Yogeshs-MacBook-Air CI_Cd % git status
On branch master_1
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   backend/api.json

no changes added to commit (use "git add" and/or "git commit -a")
yogesh@Yogeshs-MacBook-Air CI_Cd % git add .
yogesh@Yogeshs-MacBook-Air CI_Cd % git commit -m "commit form branch master_1"
[master_1 f60cddb] commit form branch master_1
 Committer: Yogesh <yogesh@Yogeshs-MacBook-Air.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit
```

Branch list, git status and commit on master_1

13. Backend API in master_2: Create the /submittodoitem route that accepts itemName and itemDescription via POST and stores them in MongoDB.

```
yogesh@Yogeshs-MacBook-Air CI_Cd % git switch  master_2
Switched to branch 'master_2'
yogesh@Yogeshs-MacBook-Air CI_Cd % git add .
yogesh@Yogeshs-MacBook-Air CI_Cd % git commit -m "commit form master_2"
[master_2 1f1f511] commit form master_2
 Committer: Yogesh <yogesh@Yogeshs-MacBook-Air.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 2 insertions(+), 1 deletion(-)
```

Switching to master_2, add and commit

14. Merging Changes: Merge master_1 and master_2 into main.

```
 1 file changed, 2 insertions(+), 1 deletion(-)
yogesh@Yogeshs-MacBook-Air CI_Cd % git switch main
Switched to branch 'main'
yogesh@Yogeshs-MacBook-Air CI_Cd % git merge --no-ff master_1
Merge made by the 'ort' strategy.
 backend/api.json | 3 ++-
 1 file changed, 2 insertions(+), 1 deletion(-)
yogesh@Yogeshs-MacBook-Air CI_Cd % git merge --no-ff master_2
Auto-merging backend/api.json
CONFLICT (content): Merge conflict in backend/api.json
Automatic merge failed; fix conflicts and then commit the result.
yogesh@Yogeshs-MacBook-Air CI_Cd % git commit
[main 8b9303f] Merge branch 'master_2'
 Committer: Yogesh <yogesh@Yogeshs-MacBook-Air.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

yogesh@Yogeshs-MacBook-Air CI_Cd %
```

Merging master_1, then master_2 (conflict in backend/api.json) and merge commit

Task 4: Sequential Commits, Reset & Rebase

15. Enhancing the To-Do Form in master_1: Add the fields Item ID, Item UUID, Item Hash.

```
git switch master_1
# edit the To-Do form: add Item ID, Item UUID, Item Hash fields
```

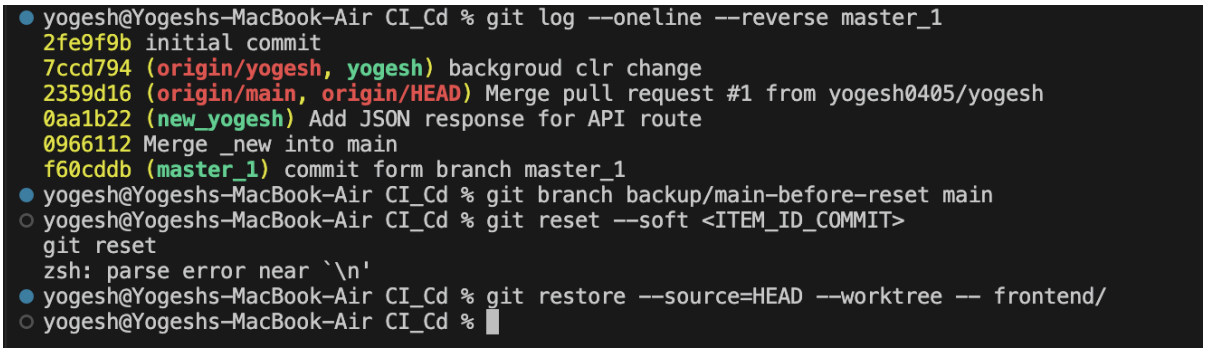
16. Committing in Sequence: Commit each field separately – Item ID, then Item UUID, then Item Hash.

```
git add frontend/
git commit -m "Add Item ID field"
git add frontend/
git commit -m "Add Item UUID field"
git add frontend/
git commit -m "Add Item Hash field"
git log --oneline
```

17. Merging to main: Merge master_1 into main.

```
git switch main
git merge --no-ff master_1
```

18. Git Reset: In main, roll back to the commit where only Item ID was added using git reset --soft, re-commit and merge to main.



Backup branch and git reset --soft attempt

Commands for the reset:

```
git log --oneline
git reset --soft master_1~2
git commit -m "Re-commit Item ID state"
git merge --no-ff master_1
```

19. Rebasing: Rebase main onto master_1 using git rebase main master_1, preserving individual commits (no squash).

```
git switch master_1
git rebase main master_1
git log --oneline --graph
```